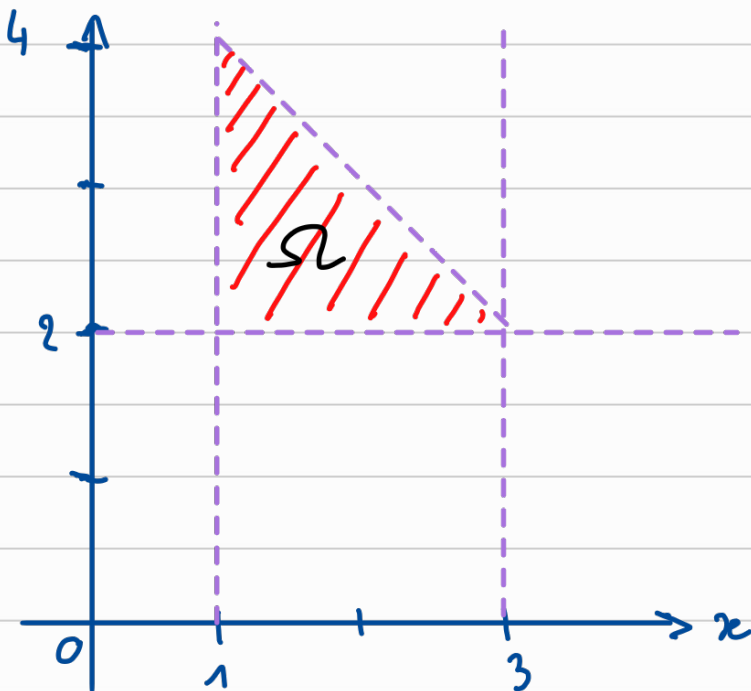


Exo 1:

• $I = \int_{\Omega} \frac{dx dy}{(x+y)^3}$ où $\Omega = \{(x,y) \in \mathbb{R}^2 \mid \begin{cases} 1 < x < 3 \\ y > 2 \\ x+y < 5 \end{cases}\}$



$f: \begin{cases} (\mathbb{R}_+^*)^2 \rightarrow \mathbb{R} \\ (x,y) \mapsto \frac{1}{(x+y)^3} \end{cases}$ est bien définie, positive et continue
donc bornée sur $\bar{\Omega}$, compact.

$\Rightarrow \int_{\Omega} f(x,y) dx dy$ est bien définie (et finie car $\mu_f(\Omega) < +\infty$)
 $\hookrightarrow$ (surface finie)

Rappel: intégrale produit / Fubini (cf. ch 4)

$$\int_{\mathbb{R}^2} |f(x,y)| dx dy < +\infty$$

$$\begin{aligned} \Rightarrow \int_{\mathbb{R}^2} f(x,y) dx dy &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y) dx \right) dy \\ &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y) dy \right) dx \end{aligned}$$

Ici, $\int_{\Omega} f(x,y) dx dy$

$$= \int_{\mathbb{R}^2} f(x,y) \chi_{\Omega}(x,y) dx dy$$

$$= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y) \chi_{\Omega}(x,y) dx \right) dy \quad (1)$$

$$= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} f(x,y) \chi_{\Omega}(x,y) dy \right) dx \quad (2)$$

$$(1) \int_2^4 \left(\int_1^{5-y} \frac{dx}{(x+y)^3} \right) dy = \int_2^4 \left[-\frac{1}{2(x+y)^2} \right]_1^{5-y} dy$$

$$= \frac{1}{2} \int_2^4 \left(\frac{1}{(y+1)^2} - \frac{1}{25} \right) dy$$

$$= \frac{1}{2} \left[-\frac{1}{y+1} - \frac{y}{25} \right]_2^4$$

$$= \frac{1}{2} \left(-\frac{1}{5} - \frac{4}{25} + \frac{1}{3} + \frac{2}{25} \right) = \frac{1}{2} \left(-\frac{15}{75} - \frac{12}{75} + \frac{25}{75} + \frac{6}{75} \right)$$

$$= \frac{1}{2} \times \frac{4}{75} = \frac{2}{75} \geq 0$$

Vérif:

$$(2) \int_1^3 \left(\int_2^{5-x} \frac{dy}{(x+y)^3} \right) dx = \int_1^3 \left[-\frac{1}{2(x+y)^2} \right]_2^{5-x} dx$$

$$= \frac{1}{2} \int_1^3 \left(\frac{1}{(x+2)^2} - \frac{1}{25} \right) dx$$

$$= \frac{1}{2} \left[-\frac{1}{x+2} - \frac{x}{25} \right]_1^3$$

$$= \frac{1}{2} \left(-\frac{1}{5} - \frac{3}{25} + \frac{1}{3} + \frac{1}{25} \right) = \frac{1}{2} \left(-\frac{15}{75} - \frac{9}{75} + \frac{25}{75} + \frac{3}{75} \right)$$

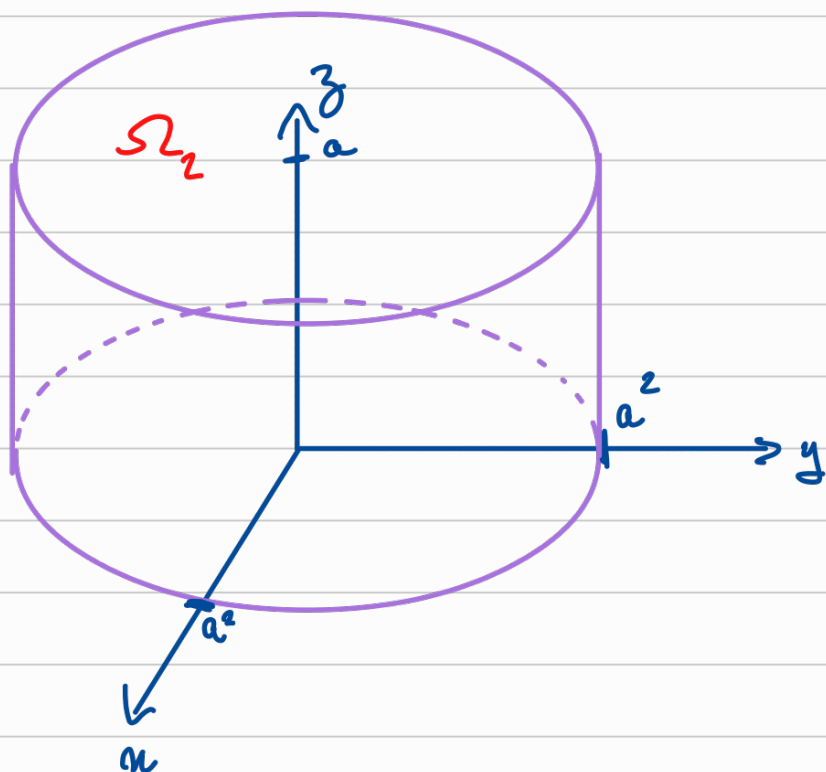
$$= \frac{1}{2} \times \frac{4}{75} = \frac{2}{75} \geq 0$$

$$\Rightarrow \boxed{I = \frac{1}{75}}$$

Exo 2:

$$\bullet \quad I = \int_{\Omega_2} \frac{z \, dx \, dy \, dz}{\sqrt{x^2 + y^2}} \quad \text{où } \Omega_2 = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \begin{cases} x^2 + y^2 < a^2 \\ 0 < z \leq a \end{cases} \right\}$$

$$f: \begin{cases} \mathbb{R}^3 \rightarrow \mathbb{R} \\ (x, y, z) \mapsto \frac{z}{\sqrt{x^2 + y^2}} \end{cases} \quad \text{est positive sur } \Omega_2$$



$$\mu_1(\Omega_2) < +\infty \\ (\text{car } \Omega_2 \text{ est borné})$$

$$\text{Soit } \varepsilon > 0, \text{ on pose } \Omega_\varepsilon = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \begin{cases} \varepsilon^2 < x^2 + y^2 < a^2 \\ 0 < z \leq a \end{cases} \right\}$$

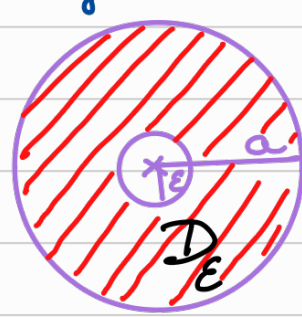
de sorte que f soit continue et $\geq c$ sur Ω_ε , borné et fermé (ie compact).

$$(\text{comme } \mu_1(\Omega_\varepsilon) < +\infty \text{ (car } \Omega_\varepsilon \text{ borné)})$$

$$\Rightarrow \int_{\Omega_\varepsilon} f(x, y, z) \, dx \, dy \, dz \text{ est bien définie } (f \geq 0) \text{ et } < +\infty.$$

Par Fubini,

$$I_\varepsilon = \int_{\Omega_\varepsilon} f(x, y, z) \, dx \, dy \, dz = \int_0^a z \left(\int_{D_\varepsilon} \frac{dx \, dy}{\sqrt{x^2 + y^2}} \right) dz$$



Calculons $I_{D_\epsilon} = \int_{D_\epsilon} \frac{dx dy}{\sqrt{x^2 + y^2}}$ en passant en polaires

Rappel: Formule de changement de variable

$$\int_Y f(y) dy = \int_X f(\varphi(x)) |\det \varphi'(x)| dx$$

$\downarrow$ $dy_1 \dots dy_n$ $\downarrow$ $dx_1 \dots dx_n$

où $\varphi: \begin{cases} X \subset \mathbb{K}^n \rightarrow Y \subset \mathbb{K}^n \\ x \mapsto \varphi(x) = y \end{cases}$

avec φ bijective, φ $\mathcal{C}^1$ et φ^{-1} $\mathcal{C}^1$ (φ $\mathcal{C}^1$ -difféomorphisme)

On pose alors $\varphi: \begin{cases}]\epsilon, a[\times]0, 2\pi[\rightarrow D_\epsilon \\ (p, \theta) \mapsto (p \cos(\theta), p \sin(\theta)) \end{cases}$

$= \begin{pmatrix} p \cos(\theta) \\ p \sin(\theta) \end{pmatrix} \rightarrow \begin{matrix} \varphi_1(p, \theta) \\ \varphi_2(p, \theta) \end{matrix}$

$\varphi'(p, \theta) = \begin{bmatrix} \frac{\partial \varphi}{\partial p}(p, \theta) & \frac{\partial \varphi}{\partial \theta}(p, \theta) \end{bmatrix} \in \mathcal{M}_2(\mathbb{R})$

$$= \begin{pmatrix} \frac{\partial \varphi_1}{\partial p}(p, \theta) & \frac{\partial \varphi_1}{\partial \theta}(p, \theta) \\ \frac{\partial \varphi_2}{\partial p}(p, \theta) & \frac{\partial \varphi_2}{\partial \theta}(p, \theta) \end{pmatrix} = \begin{pmatrix} \cos(\theta) & -p \sin(\theta) \\ \sin(\theta) & p \cos(\theta) \end{pmatrix}$$

$$\Rightarrow |\det(\varphi'(p, \theta))| = |p \cos(\theta)^2 + p \sin(\theta)^2| = |p| = p \quad (p \in]\epsilon, a[)$$

$$\Rightarrow I_{D_\epsilon} = \int_{D_\epsilon} \frac{dx dy}{\sqrt{x^2 + y^2}} = \int_{\substack{]\epsilon, a[\\ \times]0, 2\pi[}} \frac{p \, dp \, d\theta}{\sqrt{p^2 \cos(\theta)^2 + p^2 \sin(\theta)^2}} = \int_{]\epsilon, a[\times]0, 2\pi[} dp \, d\theta = 2\pi(a - \epsilon)$$

$$\begin{aligned} \Rightarrow I_\varepsilon &= \int_0^a z \mathbb{I}_{D_\varepsilon} dz = 2\pi(a-\varepsilon) \int_0^a z dz \\ &= 2\pi(a-\varepsilon) \frac{a^2}{2} = \pi a^2(a-\varepsilon) \end{aligned}$$

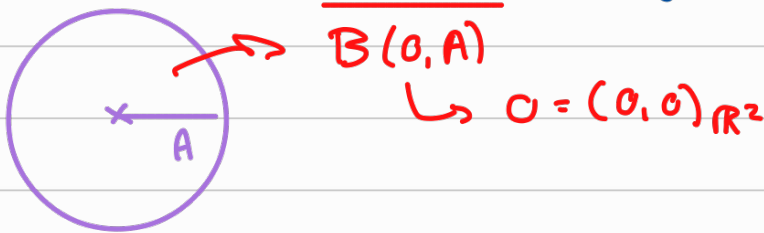
$$\Rightarrow I = \lim_{\varepsilon \rightarrow 0^+} I_\varepsilon$$

$$\Rightarrow \boxed{I = \pi a^3}$$

Exo 3:

Soit $a > 0$

$$\begin{aligned} \bullet I_a &= \int_{\mathbb{R}^2} e^{-a(x^2+y^2)} dx dy & f: \begin{cases} \mathbb{R}^2 \rightarrow \mathbb{R} \\ (x,y) \mapsto e^{-a(x^2+y^2)} \end{cases} \\ & & f \geq 0 \text{ sur } \mathbb{R}^2. \end{aligned}$$



$$\begin{aligned} \text{On calcule } \int_{B(0,A)} e^{-a(x^2+y^2)} dx dy &= \int e^{-a\rho^2} \rho d\rho d\theta \quad (\text{même changement de variable}) \\ &= \int_0^{2\pi} \left(\int_0^A e^{-a\rho^2} \rho d\rho \right) d\theta = \int_0^{2\pi} \left[\frac{e^{-a\rho^2}}{-2a} \right]_0^A d\theta \\ &= \int_0^{2\pi} \left(\frac{e^{-aA^2}}{-2a} + \frac{1}{2a} \right) d\theta = \pi \frac{1 - e^{-aA^2}}{a} \xrightarrow{A \rightarrow +\infty} \frac{\pi}{a} \end{aligned}$$

$$\Rightarrow \boxed{I_a = \frac{\pi}{a}}$$

En particulier, pour $a=1$,

$$I_1 = \pi = \int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy = \int_{\mathbb{R}^2} e^{-x^2} e^{-y^2} dx dy$$

$$\begin{aligned}
 \text{Fubini} \Rightarrow \pi &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} e^{-x^2} e^{-y^2} dx \right) dy \\
 &= \int_{\mathbb{R}} e^{-y^2} \left(\int_{\mathbb{R}} e^{-x^2} dx \right) dy \\
 &= \left(\int_{\mathbb{R}} e^{-x^2} dx \right) \times \left(\int_{\mathbb{R}} e^{-y^2} dy \right) \quad \downarrow \text{variables muettes} \\
 &= \left(\int_{\mathbb{R}} e^{-x^2} dx \right)^2
 \end{aligned}$$

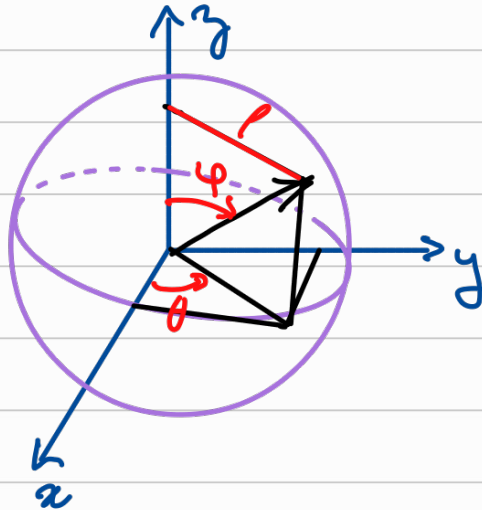
$$\Rightarrow \int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$$

Exo 4:

$$I = \int_{\Omega} \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}} \quad \text{où } \Omega = \{(x, y, z) \in \mathbb{R}^3 \mid 1 < x^2 + y^2 + z^2 < 4\}$$

On passe en sphériques :

$$\begin{cases}
 z = \rho \cos(\varphi) \\
 x = \rho \sin(\varphi) \cos(\theta) \\
 y = \rho \sin(\varphi) \sin(\theta)
 \end{cases}$$



$$\begin{aligned}
 \psi: & \begin{cases}]1, 2[\times]0, 2\pi[\times]0, \pi[\rightarrow \Omega \\ (\rho, \theta, \varphi) \end{cases} \mapsto (x, y, z) = \begin{pmatrix} \rho \sin(\varphi) \cos(\theta) \\ \rho \sin(\varphi) \sin(\theta) \\ \rho \cos(\varphi) \end{pmatrix}
 \end{aligned}$$

$$\theta \in [0, 2\pi], \varphi \in [0, \pi], \rho \in \mathbb{R}_+.$$

$$\psi'(\rho, \theta, \varphi) = \begin{pmatrix} \frac{\partial \psi}{\partial \rho} & \frac{\partial \psi}{\partial \theta} & \frac{\partial \psi}{\partial \varphi} \end{pmatrix} (\rho, \theta, \varphi)$$

$$= \begin{pmatrix} \sin(\varphi) \cos(\theta) & -\rho \sin(\varphi) \sin(\theta) & \rho \cos(\varphi) \cos(\theta) \\ \sin(\varphi) \sin(\theta) & \rho \sin(\varphi) \cos(\theta) & \rho \cos(\varphi) \sin(\theta) \\ \cos(\varphi) & 0 & -\rho \sin(\varphi) \end{pmatrix}$$

$$\Rightarrow |\det \Psi'(\rho, \theta, \varphi)| = \left| \cos(\varphi) \left(-\rho^2 \sin(\varphi) \cos(\varphi) \sin(\theta)^2 - \rho^2 \sin(\varphi) \cos(\varphi) \cos(\theta)^2 \right) \right. \\ \left. - \rho \sin(\varphi) \left(\rho \sin(\varphi)^2 \cos(\theta)^2 + \rho \sin(\varphi)^2 \sin(\theta)^2 \right) \right|$$

3^e ligne ^{der.} ↗

$$= \left| -\rho^2 \sin(\varphi) \cos(\varphi)^2 - \rho^2 \sin(\varphi)^3 \right|$$

$$= \left| -\rho^2 \sin(\varphi) (\cos(\varphi)^2 + \sin(\varphi)^2) \right| = \left| -\rho^2 \sin(\varphi) \right|$$

$$\Rightarrow I = \int_1^2 \left(\int_0^{2\pi} \left(\int_0^\pi \frac{\rho^2 \sin(\varphi) d\varphi}{\sqrt{x^2 + y^2 + z^2}} \right) d\theta \right) d\rho$$

(car $\varphi \in [0, \pi] \Rightarrow \sin(\varphi) \geq 0$)

$$\text{Or, } \sqrt{x^2 + y^2 + z^2} = \sqrt{\rho^2 (\sin(\varphi)^2 \cos(\theta)^2 + \sin(\varphi)^2 \sin(\theta)^2 + \cos(\varphi)^2)} \\ = \rho \sqrt{\sin(\varphi)^2 + \cos(\varphi)^2} = \rho$$

$$\Rightarrow I = \int_1^2 \left(\int_0^{2\pi} \left(\rho \int_0^\pi \sin(\varphi) d\varphi \right) d\theta \right) d\rho$$

$$= \int_1^2 \left(\int_0^{2\pi} \rho [-\cos(\varphi)]_0^\pi d\theta \right) d\rho$$

$$= \int_1^2 \left(2\rho \int_0^{2\pi} d\theta \right) d\rho = \int_1^2 4\pi \rho d\rho = 4\pi \left[\frac{\rho^2}{2} \right]_1^2$$

$$= 4\pi \left(2 - \frac{1}{2}\right)$$

$$\Rightarrow \boxed{I = 6\pi}$$

